Steering Troubleshoot + Production Servo Install

AQUABOTS — Steering Troubleshoot + Production Servo Install

Date: 2026-04-23 | **Owner:** John | **Team 26061**

Goal: Diagnose CCW-circling issue, swap in production servo + new BEC, recalibrate, bench-verify before next water test.

Tools needed: QGC (laptop), USB-to-Pixhawk cable, protractor or angle gauge, phone camera, Phillips/hex drivers, soldering iron (if BEC leads need redoing), multimeter, zip ties.

Prereqs before starting: Boat on bench, hull open, battery DISCONNECTED until step 5.

PART A — Diagnostic on Current Servo (do BEFORE swapping)

Purpose: rule out trim/parameter issues vs. pure torque. If the current servo can't hit full right travel on the bench with no water load, it's not a torque problem — it's a parameter problem, and swapping to the bigger servo alone won't fix it.

- [ ] **A1.** Boat out of water. Battery connected. Pixhawk powered, RC on.
- [ ] **A2.** In QGC → Parameters, note current values (write down):
  - [ ] `SERVO1_FUNCTION = ___` *(should be 26 = GroundSteering)*
  - [ ] `SERVO1_MIN = ___`
  - [ ] `SERVO1_MAX = ___`
  - [ ] `SERVO1_TRIM = ___`
  - [ ] `SERVO1_REVERSED = ___`
  - [ ] `RC1_MIN / MAX / TRIM = ___ / ___ / ___`
- [ ] **A3.** With RC stick centered, observe rudder. Is it mechanically straight with hull centerline?
  - PASS: rudder straight → trim OK
  - FAIL: rudder deflected → **this alone could explain the CCW circles.** Note direction and rough angle.

A4. RC stick hard LEFT. Observe rudder. Measure max deflection angle from center. **Angle = ___°**

- [ ]

A5. RC stick hard RIGHT. Measure max deflection angle from center. **Angle = ___°**

- [ ]

A6. Compare A4 vs A5:

- Symmetric (within ~3°) → mechanical travel OK
- Asymmetric → servo hitting mechanical stop one side before reaching SERVO1_MAX, **or** linkage geometry off, **or** TRIM is not at mechanical center

- [ ]

A7. Stall test: with power on and stick centered, apply moderate hand pressure to the rudder blade trying to push it off center.

- Holds firm → torque adequate for bench (water load is higher but this tells you the servo isn't back-driving at rest)
- Back-drives easily → torque is marginal; production servo will help

- [ ]

A8. Record findings. If A3/A6 show asymmetry or off-center → **primary fix is calibration, not hardware.** If A3/A6 are clean and only A7 looks weak → confirms Robert's torque hypothesis.

PART B — Power Down + Swap to Production Servo

- [ ]

B1. Disarm. Power off Pixhawk. **Disconnect main battery at kill switch.** Verify no LEDs on any powered component.

- [ ]

B2. Unplug old servo 3-pin from BEC/Pixhawk output.

- [ ]

B3. Remove old servo from hull mount. Note linkage length (measure pushrod center-to-center, write down: ___ **mm**).

- [ ]

B4. Disconnect old BEC. Install new BEC per wiring guide (P15/P16 branch, ATO 2A slow-blow fuse, 12 AWG from busbar).

- [ ] Verify fuse intact (2A SB)
- [ ] Polarity correct (RED to +, BLACK to −)
- [ ] Output leads accessible for servo plug

- [ ]

- [ ] **B5.** Install production servo (DSServo). Dry-fit first — confirm it sits where old one did, wires reach, servo arm clears hull.
- [ ] **B6.** Reattach pushrod/linkage. **Before final tightening: manually center the rudder by eye, and set the servo horn to its visual center position on the spline.** Connect linkage at this zero state.
- [ ] **B7.** Connect servo 3-pin to BEC output (signal wire to Pixhawk MAIN OUT 1, power/ground to BEC).
- [ ] **B8.** Double-check: no pinched wires, no shorts, linkage moves freely through full arc without binding.

PART C — Power Up + Parameter Baseline

- [ ] **C1.** Reconnect battery at kill switch. Pixhawk boots. Connect QGC.
- [ ] **C2.** Set safe baseline parameters for the new servo:
  - [ ] `SERVO1_FUNCTION = 26` (GroundSteering)
  - [ ] `SERVO1_MIN = 1100`
  - [ ] `SERVO1_MAX = 1900`
  - [ ] `SERVO1_TRIM = 1500`
  - [ ] `SERVO1_REVERSED = 0` (change after C5 if needed)
- [ ] **C3.** Write params, reboot Pixhawk (cycle power or soft reboot via QGC).

PART D — Centering (This is the fix for “went in circles”)

- [ ] **D1.** Arm Pixhawk in MANUAL mode. RC stick centered. Servo sits at TRIM.
- [ ] **D2.** Eyeball rudder vs. hull centerline.
  - **Straight?** → centering is good. Skip to D4.
  - **Deflected left?** → increase `SERVO1_TRIM` in small steps (e.g. +10µs) until straight.
  - **Deflected right?** → decrease `SERVO1_TRIM` in small steps until straight.

- [ ] **D3.** Final `SERVO1_TRIM = ___µs`. Save.
- [ ] **D4.** If you run out of trim range before rudder is centered (>~1600 or <~1400), **stop**. The horn is on the wrong spline tooth. Disarm, power off, pop horn one tooth, re-center mechanically, repeat C2–D3.

PART E — Travel + Reversal Check

- [ ] **E1.** Stick hard LEFT. Rudder deflects LEFT? (boat's perspective)
  - YES → reversal correct
  - NO → set `SERVO1_REVERSED = 1`, save, reboot, recheck
- [ ] **E2.** Measure deflection:
  - Hard LEFT: **___°** from center
  - Hard RIGHT: **___°** from center
- [ ] **E3.** Target: symmetric, ≥25° each side (Robert suggested +15° from whatever current max is — don't exceed ~35° or rudder will stall hydrodynamically).
- [ ] **E4.** If asymmetric: adjust `SERVO1_MIN` and `SERVO1_MAX` to equalize. Do NOT go past the physical mechanical stops.
- [ ] **E5.** Final values:
  - `SERVO1_MIN = ___` | `SERVO1_MAX = ___` | `SERVO1_TRIM = ___` | `SERVO1_REVERSED = ___`

PART F — RC Calibration + Dropout Investigation

Robert noted RC was disconnecting after a few minutes. Address this before next water test.

- [ ] **F1.** Run QGC Radio calibration (stick sweep procedure). Save.
- [ ] **F2.** Check RC failsafe settings: `FS_THR_ENABLE`, `FS_ACTION`. Confirm behavior on signal loss is HOLD or RTL, not continued last command.
- [ ] **F3.** Verify new BEC can supply servo peak current without sagging Pixhawk rail — measure BEC output voltage while working the stick rapidly. Should stay within 5% of nominal.

- [ ] **F4.** Physically inspect RC receiver: antennas free and perpendicular, not near ESC/motor wiring, no kinks.
- [ ] **F5.** Log test: power on, let sit 10 min with RC active, confirm no dropouts on bench before re-attempting water test.

PART G — Compass Calibration (do only if auto mode will be tested)

- [ ] **G1.** QGC → Sensors → Compass calibration. Follow on-screen rotation prompts.
- [ ] **G2.** Accelerometer calibration (same menu).
- [ ] **G3.** Do this with hull fully assembled and electronics in final positions. Redo after any significant component relocation.

Note: compass cal will need to be redone AGAIN after final EMI install and once hull is fully buttoned up. This round is to support bench checks only.

PART H — Final Bench Verification (before water)

- [ ] **H1.** Manual mode, throttle disabled. Stick full sweep — rudder tracks smoothly, returns to center cleanly.
- [ ] **H2.** Enable throttle (prop removed or boat blocked). Confirm throttle response is bidirectional, stops at stick center.
- [ ] **H3.** Save a parameter file from QGC as a backup snapshot: `aquabots_params_post_servo_swap_2026-04-23.params`.
- [ ] **H4.** Log final SERVO1 values above in `AQUABOTS_POWER_SYSTEM_CHANGELOG.md` and `AQUABOTS_WIRING_GUIDE.md` (new rev note).

DO-NOT-DO LIST

- Do NOT test auto mode in water until compass is re-calibrated AND EMI filters are installed. Motor EMF into an uncalibrated compass will give garbage headings and you'll be back here next week.
- Do NOT push servo travel past mechanical stops to get more deflection. You'll strip gears.
- Do NOT assume the circling was purely servo torque until you've confirmed TRIM is at mechanical center. Robert's probably right, but confirm before it's the only theory on the table.

Known Issues Still Open After This Session

- Compass + accel calibration (pending final hull close-out)
- EMI filtering install (separate checklist)
- Pool test in AUTO mode
- RC dropout root cause — may need telemetry capture if F2–F5 don't resolve